

An Extremely Projective Normed Space That Is Not Isometric to Any $\ell_1^0(M)$

Candidate counterexample packet for arXiv:1112.5750

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Status. Candidate full counterexample; likely valid, pending human review. The space $P = c_{00} + \mathbb{C}(2^{-2^j})_{j \geq 1} \subset \ell_1$ is extremely projective but is not isometrically isomorphic to $\ell_1^0(M)$ for any set M .

1 Source conjecture

In Section 3 of A. Ya. Helemskii's *Metric freedom and projectivity for classical and quantum normed modules*, arXiv:1112.5750v1 [2], the author conjectures that every extremely projective normed space is isometrically isomorphic to $\ell_1^0(M)$; see source PDF page 10 and Remark 3.6 on page 12. The author's earlier arXiv:1104.2463 states the same question and characterizes extremely projective normed spaces as near-retracts of some $\ell_1^0(\Lambda)$ [1].

3 Metrically projective normed spaces are l_1^0

Now we turn to a quite concrete question of the geometry of normed spaces. What is the structure of metrically projective spaces ?

Recall that Grothendick actually answered a similar question for *extremely* projective *Banach* spaces: they happened to be $l_1(M)$, where M is a set. It is natural to conjecture that extremely projective normed spaces are $l_1^0(M)$, but we do not know whether it is true. However, we can describe *metrically* projective spaces, non-completed and (what is easier) completed alike. This is what will be done in this section. We begin with some preparation.

We call a Banach space, say E , *metrically flat* if, for every isometry $i : F \rightarrow G$ of Banach spaces the operator $\mathbf{1}_E \widehat{\otimes} i : E \widehat{\otimes} F \rightarrow E \widehat{\otimes} G$ is also isometry.

Figure 1: The conjecture in Section 3 of the source paper (PDF page 10).

We disprove the conjecture by a one-dimensional algebraic enlargement of c_{00} inside its completion.

2 Definitions

A contractive surjection $q : Y \rightarrow X$ is *coisometric* if it maps the open unit ball of Y onto the open unit ball of X . A normed space P is *extremely projective* if, for every coisometry $q : Y \rightarrow X$,

bounded operator $\varphi : P \rightarrow X$, and $\varepsilon > 0$, there is $\psi : P \rightarrow Y$ such that

$$q\psi = \varphi, \quad \|\psi\| < \|\varphi\| + \varepsilon.$$

We write $c_{00} = \ell_1^0(\mathbb{N})$ with its usual ℓ_1 norm.

3 Statement and proof intuition

Theorem 1. *Let e_j be the canonical basis of ℓ_1 , set*

$$a_j = 2^{-2^j}, \quad x = \sum_{j=1}^{\infty} a_j e_j, \quad P = c_{00} + \mathbb{C}x \subset \ell_1,$$

and give P the inherited norm. Then P is extremely projective, but P is not isometrically isomorphic to $\ell_1^0(M)$ for any set M .

Proof intuition. The vector x is superlacunary: sufficiently far out, each tail is almost entirely its first coordinate. Replace the tail of the canonical basis by normalized successive tails of x . Tail differences recover the omitted coordinate vectors, so the new sequence is still a Hamel basis of P ; and superlacunarity makes it uniformly as close to the canonical ℓ_1 basis as desired. Hence P is an arbitrarily small distortion of c_{00} , which is enough to transfer the asymptotic lifting property defining extreme projectivity.

The space nevertheless cannot be isometric to c_{00} . Any such isometry would extend to an onto isometry of the completions ℓ_1 . Extreme points force onto ℓ_1 -isometries to permute the coordinate axes, so they preserve c_{00} and cannot map it onto the strict enlargement P .

4 Two preliminary lemmas

Lemma 2. *c_{00} is extremely projective.*

Proof. Let $q : Y \rightarrow X$ be coisometric and $\varphi : c_{00} \rightarrow X$ bounded. Given $\varepsilon > 0$, choose for every j a vector $y_j \in Y$ with $q(y_j) = \varphi(e_j)$ and $\|y_j\| < \|\varphi\| + \varepsilon/2$. This is possible by coisometry because $\|\varphi(e_j)\| \leq \|\varphi\|$. The formula $\psi(e_j) = y_j$ defines a bounded operator on c_{00} with $q\psi = \varphi$ and

$$\|\psi\| = \sup_j \|y_j\| \leq \|\varphi\| + \varepsilon/2 < \|\varphi\| + \varepsilon.$$

□

Lemma 3 (Distance-one transfer). *Suppose Q is extremely projective and, for every $\delta > 0$, there is an onto isomorphism $T : Q \rightarrow P$ with $\|T\| \|T^{-1}\| < 1 + \delta$. Then P is extremely projective.*

Proof. Rescale T so that $\|T\| = 1$. Let $q : Y \rightarrow X$ be coisometric and $\varphi : P \rightarrow X$ bounded. If $\varphi = 0$, take the zero lifting. Otherwise, given $\varepsilon > 0$, first choose T so close to isometric that $\|T^{-1}\| \|\varphi\| < \|\varphi\| + \varepsilon/2$, and then choose $\eta > 0$ with $\|T^{-1}\| \eta < \varepsilon/2$. Extreme projectivity of Q gives a lifting $\theta : Q \rightarrow Y$ of φT satisfying

$$\|\theta\| < \|\varphi T\| + \eta \leq \|\varphi\| + \eta.$$

Then $\psi = \theta T^{-1}$ lifts φ and has norm less than $\|\varphi\| + \varepsilon$.

□

5 The normalized-tail basis

For $n \geq 0$, put

$$r_n = \sum_{j>n} a_j e_j, \quad s_n = \|r_n\|_1, \quad b_n = s_n^{-1} r_n.$$

For each $N \in \mathbb{N}$, define $U_N : c_{00} \rightarrow P$ on the canonical basis by

$$U_N e_j = \begin{cases} e_j, & j \leq N, \\ b_{j-1}, & j > N. \end{cases} \quad (1)$$

Lemma 4. *Each U_N is onto, and*

$$\|U_N - I\| \leq \delta_N \longrightarrow 0.$$

Consequently

$$\|U_N\| \|U_N^{-1}\| \leq \frac{1 + \delta_N}{1 - \delta_N} \longrightarrow 1.$$

Proof. Every vector in the range of U_N belongs to P . Conversely, $r_N = s_N U_N e_{N+1}$, so

$$x = \sum_{j=1}^N a_j e_j + r_N$$

belongs to the range. For every $j > N$,

$$e_j = \frac{r_{j-1} - r_j}{a_j} = \frac{s_{j-1} U_N e_j - s_j U_N e_{j+1}}{a_j},$$

while $e_1, \dots, e_N$ are already in the range. Thus the range contains $c_{00} + \mathbb{C}x = P$.

Write $R_j = \sum_{k>j} a_k$. For $j > N$,

$$\|b_{j-1} - e_j\|_1 = \frac{2R_j}{a_j + R_j}.$$

Since $a_{j+1}/a_j = 2^{-2^j}$ and all later ratios are smaller,

$$\frac{R_j}{a_j} \leq \frac{2^{-2^j}}{1 - 2^{-2^{j+1}}} \longrightarrow 0.$$

Therefore

$$\delta_N := \sup_{j>N} \|b_{j-1} - e_j\|_1 \longrightarrow 0.$$

For a finitely supported scalar sequence (λ_j) ,

$$\|(U_N - I) \sum_j \lambda_j e_j\|_1 \leq \delta_N \sum_j |\lambda_j|.$$

Hence $\|U_N - I\| \leq \delta_N$. For large N , $\delta_N < 1$, and

$$(1 - \delta_N) \|z\|_1 \leq \|U_N z\|_1 \leq (1 + \delta_N) \|z\|_1.$$

Together with surjectivity, this gives the asserted distortion bound. $\square$

6 Proof of the counterexample

Proof of Theorem 1. Lemmas 2, 3, and 4 show that P is extremely projective.

Suppose that P is isometrically isomorphic to $\ell_1^0(M)$. Since P is separable and infinite-dimensional, M must be countably infinite, so this would give an onto isometry $J : c_{00} \rightarrow P$. Both spaces are dense in ℓ_1 , hence J extends uniquely to a surjective linear isometry $\bar{J} : \ell_1 \rightarrow \ell_1$.

The extreme points of the closed unit ball of complex ℓ_1 are precisely $\{\lambda e_j : |\lambda| = 1, j \in \mathbb{N}\}$. (If a unit vector has two nonzero coordinates, a small mass can be shifted between them to write it as a nontrivial midpoint.) Thus $\bar{J}$ permutes the coordinate axes up to unimodular scalars and therefore $\bar{J}(c_{00}) = c_{00}$. On the other hand, $\bar{J}(c_{00}) = J(c_{00}) = P$. This is impossible because $x \in P \setminus c_{00}$. $\square$

Corollary 5. *Extreme projectivity does not coincide with metric projectivity (or metric freedom) for normed spaces.*

Proof. Theorem 3.5 of [2] says that every metrically projective normed space is isometric to some $\ell_1^0(M)$. Theorem 1 therefore gives an extremely projective normed space that is not metrically projective. $\square$

7 Verification, scope, and novelty

The proof is self-contained apart from the last corollary's use of the source paper's Theorem 3.5. The included exact-arithmetic script checks finite truncations of the tail vectors, the formula for their distance to the canonical basis, the triangular invertibility, and the decay of the error. These computations are sanity checks, not substitutes for the limiting proof.

On 11 August 2026, the run indexes had no result for arXiv:1112.5750 or the conjecture. Exact-phrase and close-keyword searches of the local parsed arXiv corpus found the question only in arXiv:1104.2463 and 1112.5750. A local later citation, arXiv:1309.4974, does not discuss the conjecture. Bounded external searches returned no usable records in this environment, so novelty confidence is moderate.

Human-review recommendation. Verify the range computation in Lemma 4, the distortion transfer in Lemma 3, and the completion-isometry obstruction. No unproved dependency remains.

References

- [1] A. Ya. Helemskii, *Extreme version of projectivity for normed modules over sequence algebras*, arXiv:1104.2463v1 (2011).
- [2] A. Ya. Helemskii, *Metric freedom and projectivity for classical and quantum normed modules*, arXiv:1112.5750v1 (2011).